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Rudradev Arya
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Education

MIT-ADT, Pune **Nov. 2021 – June 2025**
B.Tech in Computer Science and Engineering, Specializing in Cloud Computing *Pune, MH*

Experience

Siemens Digital Industries Software **August 2025 – Present**
Software Engineer *Pune, MH*

- Built the **TCX Sandbox** Onboarding Portal (React, Express, MongoDB): My Environments, admin All Environments and User Management, GitLab-style search/filter/sort, config import, one-time scheduled deploys, and **SAM 2.0** in the Ansible payload
- Hardened portal APIs with **JWT**/admin middleware, webhook and Azure-creds token checks, source restrictions, and **WSS** for the in-browser shell; surfaced Vault-backed TCX passwords in the UI
- Implemented **Coverity** static analysis and **GitLab CI** pipeline rules for the sandbox, including backend defect remediation and scan-job gating
- Triaged and resolved security issues in **TCX applications** and **RHEL UBI/AMI** images; automated Polarion SI/CVE analysis with NVD, Red Hat errata, and SBOM checks (dry-run vs apply)
- Built a **GitLab CI** pipeline (**Ansible + Terraform**) that provisions throwaway AWS Windows infra, installs Teamcenter X tools and dependencies, and captures a **KMS-encrypted golden AMI**

Pratiti Technologies **July 2025 – August 2025**
DevOps Intern *Pune, MH*

- Authored **Terraform** modules to provision AWS networking, compute, and IAM, using remote state and environment-specific variables
- Wrote **Ansible** playbooks for configuration management, package/runtime installs, and repeatable Linux host setup
- Integrated Terraform and Ansible into **CI/CD** so infra and config changes applied consistently across environments

Unizoy **April 2025 – July 2025**
Backend Developer Intern *Remote*

- Built NestJS APIs for **iDelta Books**, a multi-role trading back-office that ingests daily NSE/BSE/MCX broker CSVs and computes P&L, margin, interest, and ledgers scoped by Unique Client Code and trader families
- Designed a generic async **CSV/Excel export** pipeline (BullMQ + Strategy) streaming reports to S3 with signed URLs, progress, retries, and cleanup across P&L, margin, interest, ledgers, and admin dashboards
- Implemented **dual UCC** (NSE/BSE vs MCX) P&L, dashboards, and reports with RBAC, plus super-admin homepage metrics using bulk SQL so UI totals match report modules
- Shipped **cost-log and KYC** flows (JSONB prefs, TypeORM migrations, auto-create on approval, pricing PATCH, audit history) on a hexagonal NestJS core (ports/adapters, TypeORM, JWT, 2FA)

Freelance **2024 – 2025**
Software Developer *Remote*

- **Deribit execution client:** C++17 REST + TLS WebSocket client for Deribit Testnet — place/cancel/modify, order book, and positions over libcurl, plus WebSocket++/ASIO JSON-RPC auth and live book subscribe on a threaded I/O loop
- **Crypto L2 simulator feed:** asynchronous TLS WebSocket client (Boost.Beast/Asio, OpenSSL) ingesting live OKX L2 order-book ticks, with timeouts, disconnect handling, and JSON parse isolation on a dedicated `io_context` thread
- **MahaCPD:** frontend and test coverage for a portal where Maharashtra doctors log seminar/CPD hours to earn yearly points required to keep their practitioner license
- **Stealth dApp/DAO platform:** GitHub Actions and **AWS CI/CD** for decentralized apps, and Dockerized services so local and cloud deploys used the same images

Projects

LambdaMailer | *Node.js, Amazon Lambda, Amazon SES* **February 2024**

- Leveraged **AWS Beanstalk** for scalable backend deployment and **Vercel** for frontend with which users can upload a CSV with relevant target mail IDs and other information
- Acheived a success rate of **99.2%** for **50,000** emails at just **$40/month** at maximum throughput of **200 mails/sec** and at maximum capacity
- Utilized **React** and **React Router** for a user friendly interface and **OAuth 2.0** protocol for secure authentication and authorization and minimal
- Designed the project as a microservice, communicating via APIs built with **Axios** and **Express.js**

EventU | *Next.js, Appwrite, TypeScript* **August 2023**

- Engineered a complex event management system leveraging **NoSQL** databases; streamlined unstructured data processing and achieved a **40%** reduction in data retrieval time, enhancing overall event coordination efficiency
- Designed the user interface using HTML, **TailwindCSS**, and **Next.js**, enhancing the platform's usability.
- Leveraged **Appwrite** for authentication, multiple sign-in methods, and efficient storage and file management.
- Utilized Next.js, an open-source React-based framework, to achieve server-side rendering and enhanced **SEO**. Deployed the application on **Vercel's** frontend cloud infrastructure for optimal performance.

Elevator Waiting Time Estimation and Simulation | *Arduino PoC, Python, Java, C++* and **Jan 2023**

- Implemented Various **Algorithms** using **Java** like Round Robin, Up Peak, Zoning etc to effectively simulate **Scheduling Algorithms** in an elevator
- Created a **Proof Of Concept(PoC)** model using Arduino and C++ to demonstrate real world effectiveness of Scheduling algorithms and to effectively **estimate** waiting time for an elevator

Minecraft Replica using Python | *Python, Pyglet, Linear Algebra* **April 2023**

- Utilized **Pyglet** a Python multimedia library to create a **128x128** Block replica of the original Minecraft game
- Leverage basic **Linear Algebra** to understand various game development techniques such as cursor and player movement correlation
- Implemented **object-oriented programming** practices such as abstraction to abstract away user controls and player movements
- Implemented **CUDA** to parallelize the workload of the game improving the FPS from **50 FPS** on CPU to over **270 FPS** on a mid tier GPU

Technical Skills

Languages: Python, TypeScript, JavaScript, C/C++17, HTML, CSS, PL/SQL, CUDA
Developer Tools: Git, GitLab CI, GitHub Actions, Coverity, Docker, VS Code, Postman
Cloud & IaC: AWS (EC2, AMI, IAM, S3, Lambda, SES, SQS, RDS), GCP, Terraform, Ansible
Frameworks: NestJS, React, Next.js, Express.js, Node.js, BullMQ, Boost.Asio/Beast
Database Systems: PostgreSQL, TypeORM, MongoDB, Redis, Firebase
Deployment and Monitoring: AWS Elastic Beanstalk, Vercel, VPS, Heroku

Certifications

Nvidia : Fundamentals of Accelerated Computing with CUDA C/C++ LINK
AWS Cloud Quest : Cloud Practitioner
AWS Academy Graduate : AWS Academy Cloud Architecting LINK

Achievements

- Solved more than **400**+ on various coding platforms like **Codeforces** and **GeeksForGeeks**.
- Participated and **Won** in ValiCQ, university level coding competition hosted by **CodeChef-MITSOE** Chapter.
- Participated at national level **GPU** Computing **Codeathon** organized by NVIDIA DLI.
- Top 30 out of 2000+ peers on the University level GeeksForGeeks Leaderboard and Improving.